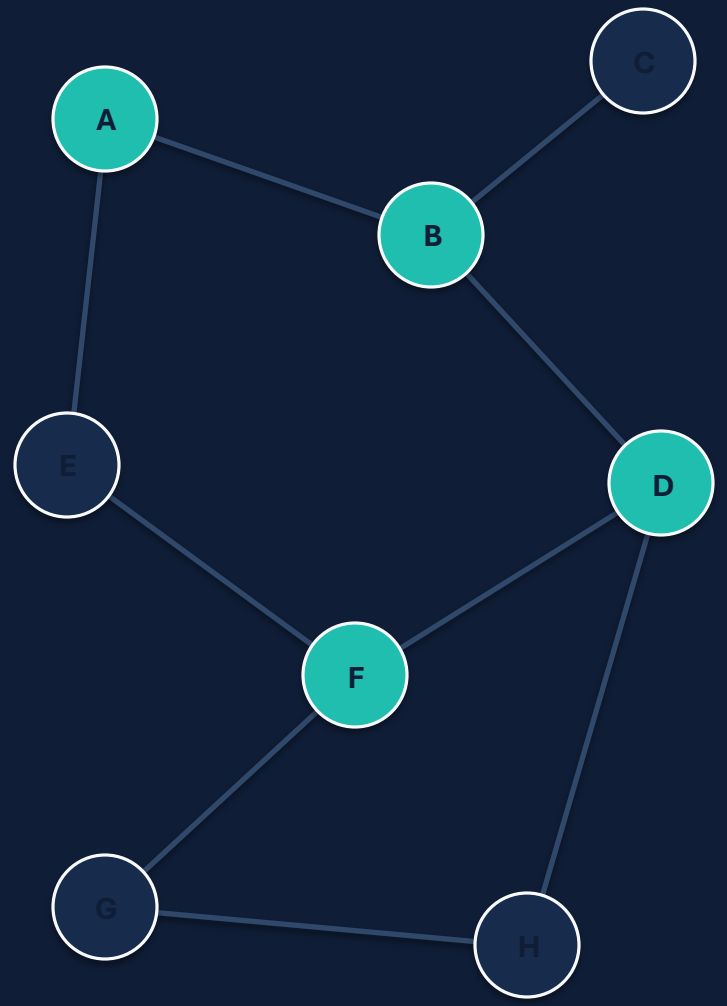
DESIGN & ANALYSIS OF ALGORITHMS

KRUSKAL'S ALGORITHM

A simple, greedy approach to finding a Minimum Spanning Tree



B.Tech • Computer Science & Engineering

Concepts | Worked example | Complexity

DAA PRESENTATION

What is Kruskal’s Algorithm?

Start with the smallest edge and keep building a low-cost tree.

CORE DEFINITION

Kruskal’s algorithm is a greedy algorithm that finds a Minimum Spanning Tree (MST) of a connected, weighted, undirected graph.

It considers edges in increasing order of weight and accepts an edge only if it does not create a cycle.

MINIMUM SPANNING TREE

A set of edges that connects every vertex, has no cycle, and has the minimum total weight.

WHY “GREEDY”?

At each step, it chooses the cheapest available edge that is safe to add.

Key idea: connect all vertices at the lowest possible total cost—without forming a loop.

How does it work?

A repeatable process: sort → inspect → accept or reject → stop.

1

Sort edges

Arrange all edges from lowest to highest weight.

2

Pick the lightest

Take the next smallest edge in the sorted list.

3

Check for a cycle

If it creates a cycle, reject it; otherwise, accept it.

4

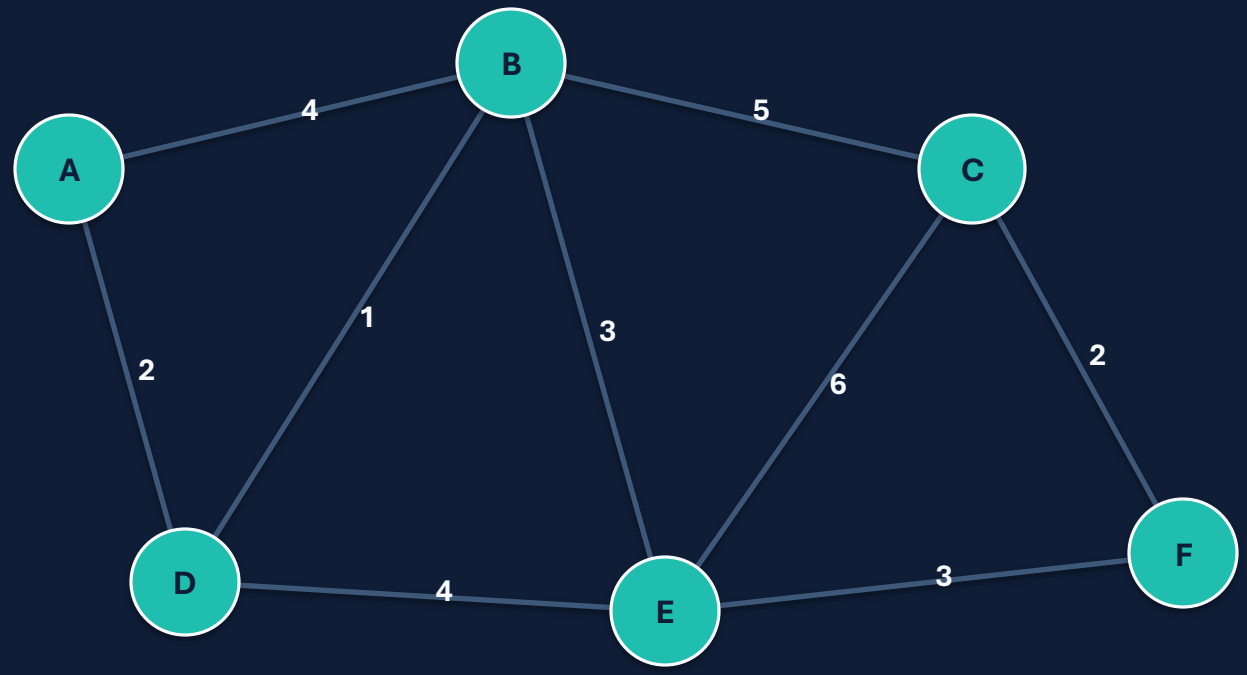
Stop at V − 1

A spanning tree with V vertices has exactly V − 1 edges.

Cycle check is the crucial rule: never add an edge that closes a loop.

Example graph: 6 vertices, 9 weighted edges

We will find the MST by examining edges from smallest to largest.



EDGE LIST (WEIGHT)

BD = 1
AD = 2 CF = 2
BE = 3 EF = 3
AB = 4 DE = 4
BC = 5 CE = 6

All weights are costs.

Goal: select 5 edges (V − 1) that connect A, B, C, D, E and F with no cycles.

Sort the edges and make safe choices

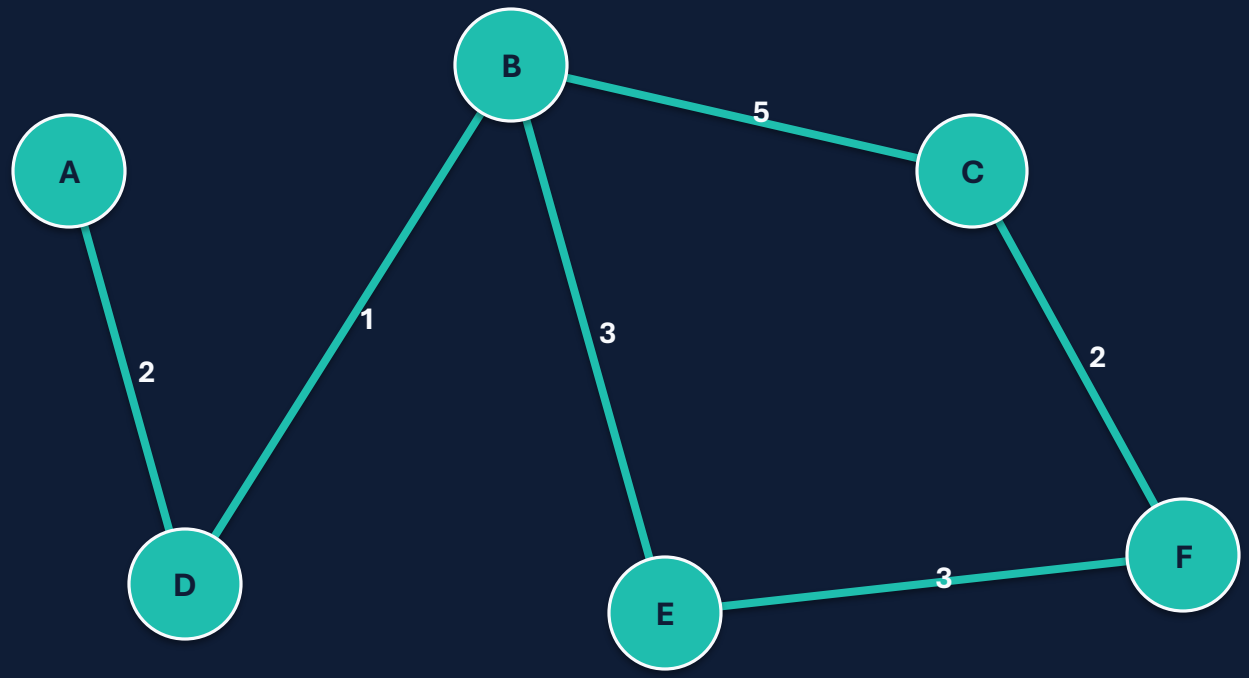
A rejected edge is one that would create a cycle.

ORDER	EDGE	WEIGHT	DECISION	REASON
1	BD	1	ACCEPT	Connects B and D
2	AD	2	ACCEPT	Adds A to the component
3	CF	2	ACCEPT	Starts the C–F component
4	BE	3	ACCEPT	Adds E to the main component
5	EF	3	ACCEPT	Joins the two components
6	AB	4	REJECT	A, B and D are already connected
7	DE	4	REJECT	Would form a cycle
8	BC	5	ACCEPT	Joins the remaining components

Stop after 5 accepted edges. The final accepted edge is BC (weight 5).

The Minimum Spanning Tree

Five accepted edges connect all six vertices; no cycle remains.



TOTAL MST WEIGHT

1 + 2 + 2 + 3 + 3 + 5

= 16

FINAL CHECK

6 vertices • 5 edges • no cycles

MST edge set: {BD, AD, CF, BE, EF, BC}

Complexity, advantages & key takeaways

Remember the greedy rule and the V − 1 stopping condition.

TIME COMPLEXITY

O(E log E)

Mainly from sorting E edges.

SPACE COMPLEXITY

O(V)

For parent / disjoint-set data.

BEST USE

Sparse graphs

Often efficient when edges are few.

ADVANTAGES

- Simple greedy strategy
- Works well with edge sorting

REMEMBER

- Accept only cycle-free edges
- Stop when V − 1 edges are accepted

Thank you • Questions?

Pseudocode: Kruskal’s Algorithm

Disjoint Set Union (DSU) helps check whether an edge would create a cycle.

```
KRUSKAL(G):
  MST = empty set
  Sort all edges by increasing weight
  For each vertex v: Make-Set(v)
  For each edge (u, v) in sorted order:
    If Find(u) ≠ Find(v):
      Add (u, v) to MST
      Union(u, v)
    If |MST| = V − 1: Stop
  Return MST
```

FIND(u)

Checks which connected component contains u.

UNION(u, v)

Combines two components after accepting an edge.

DSU makes cycle detection efficient.

Applications & when to use it

Kruskal’s algorithm connects a network at minimum total cost.

REAL-WORLD APPLICATIONS

- Designing low-cost computer networks
- Planning roads, railways, or utility lines
- Connecting buildings with cables or pipes
- Building communication networks

KRUSKAL vs. PRIM

KRUSKAL	PRIM
Chooses the globally smallest safe edge.	Grows one tree from a starting vertex.
Sorts edges first.	Chooses a cheapest outgoing edge.

Both algorithms find an MST for a connected, weighted, undirected graph.

THANK YOU

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